The Rejection Region Approach and the Z-test

Contents

1. Overview 2
2. Terms we build on 3
3. The logic of a test 3
4. The z statistic: counting standard errors 4
    4.1 The idea on the rats 4
    4.2 The formula 5
    4.3 Assumptions 6
5. How rare is too rare: the significance level 6
6. The rejection region and the critical value 8
7. One tail or two: the same data, two questions 10
8. Example: did a training program raise productivity? 10
9. Example: do chips packets weigh 50 g? 11
10. How the significance level moves the rejection region 13
11. The weakness of the rejection region approach 14
12. Summary 16
13. Sources 16
14. Key terms 17

Where this fits: Pipeline map step 0 (Foundations). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Standard normal and the z-table; Central limit theorem; Hypothesis testing: null and alternative.
- **Leads to:** One-sample proportion test.

1. Overview

Key point: In the **rejection region approach** (G-1663) we assume H_0 (the **null hypothesis** (G-1361): the claim of no effect) is true, measure how far the sample lands from what H_0 predicts, and reject H_0 when the sample lands in a tail region whose area is the **significance level** (G-1801) α.

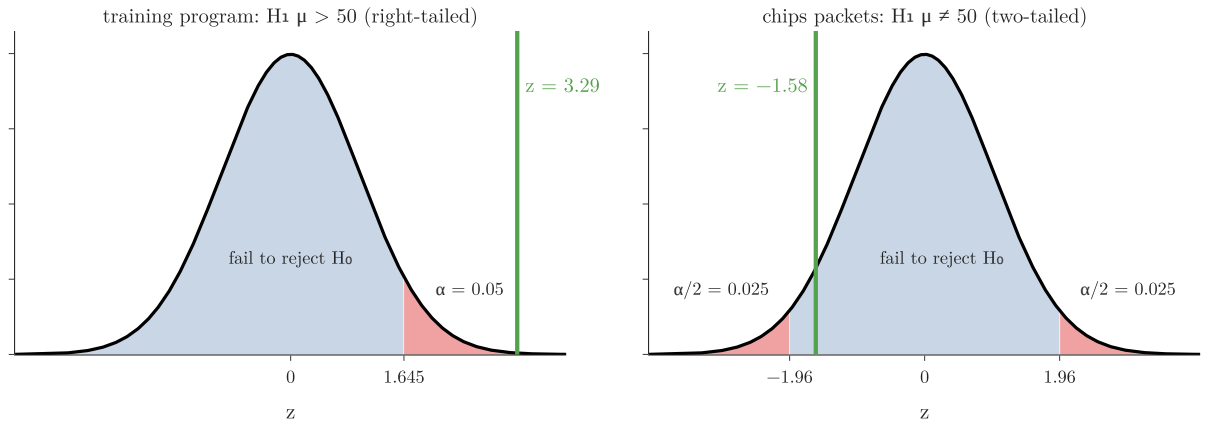


Figure 1: The factory and chips examples of this Note: the observed z (green) against the rejection region (red), $\alpha = 0.05$

The eight steps of a hypothesis test were listed after H_0 and H_1 (the **alternative hypothesis** (G-193): the claim that there is an effect) were set up. Here we run those steps end to end with the one-sample **z-test** (G-2143). Figure 1 shows where the two main examples end: the training program's z of 3.29 lands in the red rejection region, the chips' z of -1.58 does not.

We build the idea in this order:

1. the logic: assume H_0, then ask how surprising the sample is (section 3);
2. the **z statistic** (G-2137): count the distance in standard errors (section 4);
3. the significance level α: how rare is too rare (section 5);
4. the rejection region and the **critical value** (G-504) (section 6);
5. one tail or two (section 7);
6. two complete examples, right-tailed and two-tailed (sections 8 and 9);

7. how α moves the rejection region (section 10);
8. the weakness of the approach, which **p-values** (G-1433) (the tail area beyond our own sample) fix (section 11).

2. Terms we build on

> **Key point:** The standard normal curve and z-table, the central limit theorem with its standard error, and **critical values** (G-504).

- **Standard normal and z-table:** a z-score has the standard normal distribution, and the z-table gives the area to the left of any z (see the z-table).
- **Central limit theorem and standard error:** for a sample of $n \geq 30$, the sample mean $\bar{X}$ is approximately normal with mean μ and standard deviation $\sigma/\sqrt{n}$, the standard error (G-1872) (see the central limit theorem).
- **Critical value:** the z value that leaves a given area in the tail; 1.96 leaves 2.5% in each tail (see finding the critical value).

3. The logic of a test

> **Key point:** We pretend H_0 is true, work out where sample means would then land, and reject H_0 only if our sample lands somewhere very unlikely.

The question. Rats that get no drug take **1.2 seconds** on average to respond to a stimulus. We inject **100 rats** with a new drug. Their mean response time is **1.05 seconds**, with a sample standard deviation of **0.5 seconds**. Does the drug change the response time?

The hypotheses. The **null hypothesis** (G-1361) is the status quo: the drug has no effect, so the mean with the drug is still 1.2 s. The **alternative hypothesis** (G-193) is that the drug has an effect:

$$H_0 : \mu = 1.2$$

$$H_1 : \mu \neq 1.2$$

The logic, in plain words.

1. Assume H_0 is true.
2. Work out how sample means of 100 rats would spread out if that were so.
3. Ask how likely a sample mean as far from 1.2 s as ours would be.
4. If that is very unlikely, H_0 is probably false: we reject it.

Step 2 is a picture we already know. If we took many samples of 100 rats, their means would form the **sampling distribution of the sample mean** (G-1735). By the central limit theorem it is close to normal. Under H_0 its centre is 1.2 s. Its spread is the standard error, $\sigma/\sqrt{n}$.

We do not know σ, so we use the sample's $s = 0.5$ in its place. With 100 rats that is a close approximation (the next section says how close):

$$\text{standard error} \approx \frac{0.5}{\sqrt{100}} = \frac{0.5}{10} = 0.05 \text{ s}$$

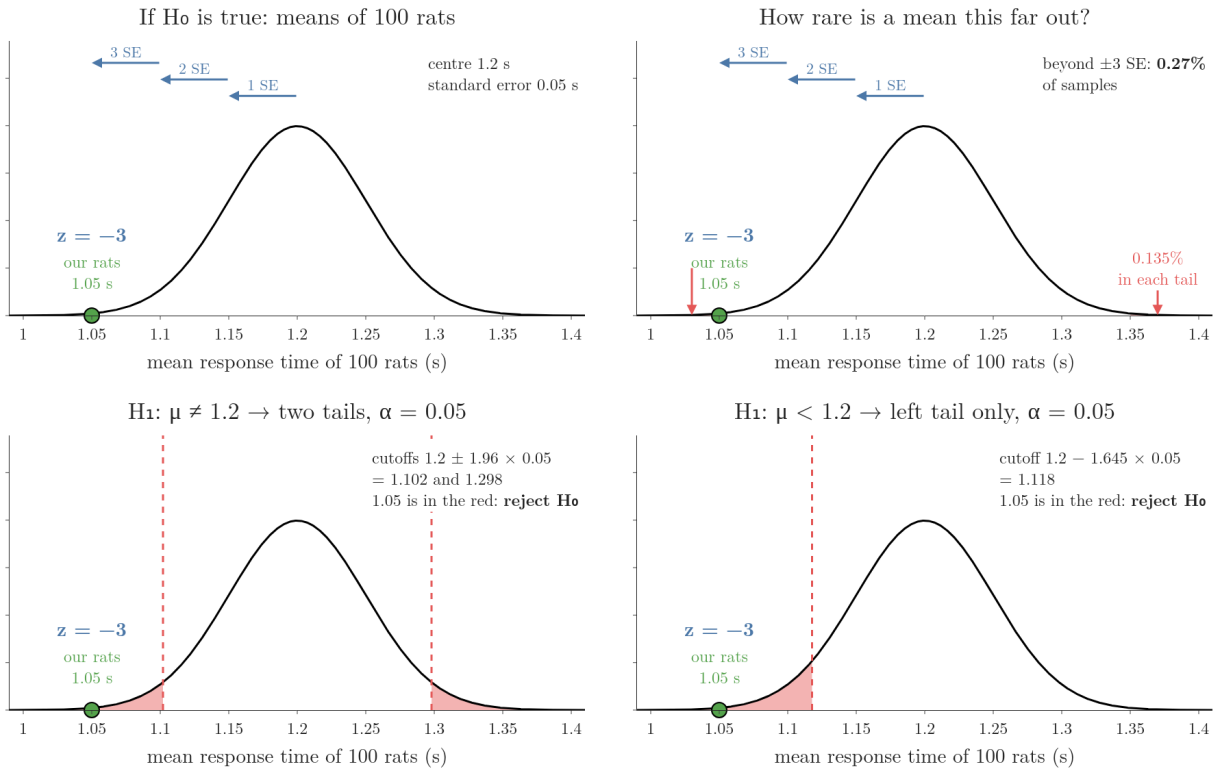


Figure 2: The rat drug test step by step: the sampling distribution of the mean if H_0 is true (centre 1.2 s, standard error 0.05 s); 1.05 s sits 3 standard errors below; only 0.27% of samples land that far out; then the two-tailed and the one-tailed rejection regions at $\alpha = 0.05$. Idea after Khan Academy, "Hypothesis testing and p-values"

In the first frames of Figure 2, watch the green dot at 1.05 s: it sits far out in the left tail of the curve that H_0 predicts.

4. The z statistic: counting standard errors

> **Key point:** The **z statistic** (G-2137) measures how many standard errors the sample mean lies from the mean claimed by H_0; the one-sample **z-test** (G-2143) needs σ and a normal sampling distribution.

4.1 The idea on the rats

How far is 1.05 s from 1.2 s? In seconds the gap is 0.15. In standard errors it is

$$\frac{1.05 - 1.2}{0.05} = \frac{-0.15}{0.05} = -3$$

In Figure 2, the blue arrows lay three standard-error steps from 1.2 s down to 1.05 s. Our sample mean is **3 standard errors below** what H_0 predicts. This count is the **z statistic** (G-2137), a **test statistic** (G-1963): the one number a test computes from the sample to make its decision.

4.2 The formula

If H_0 is true, the population mean is μ_0, the value in H_0. The sample mean then varies around μ_0 with standard error $\sigma/\sqrt{n}$, so standardizing it gives a standard normal variable.

1. **In words:** the distance between the sample mean and the mean claimed by H_0, measured in standard errors.
2. **Formula:**
$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$
where $\bar{x}$ is the sample mean, μ_0 the mean in H_0, σ the population standard deviation and n the sample size.
3. **Example:** in a car factory an employee produces $\mu_0 = 50$ units a day on average, with $\sigma = 5$; after a training program, a sample of $n = 30$ employees averages $\bar{x} = 53$ units a day:
$$z = \frac{53 - 50}{5/\sqrt{30}} = \frac{3}{0.913} = 3.29$$

We divide by $\sigma/\sqrt{n}$, not by σ, because we are standardizing a sample **mean**, whose spread is the standard error.

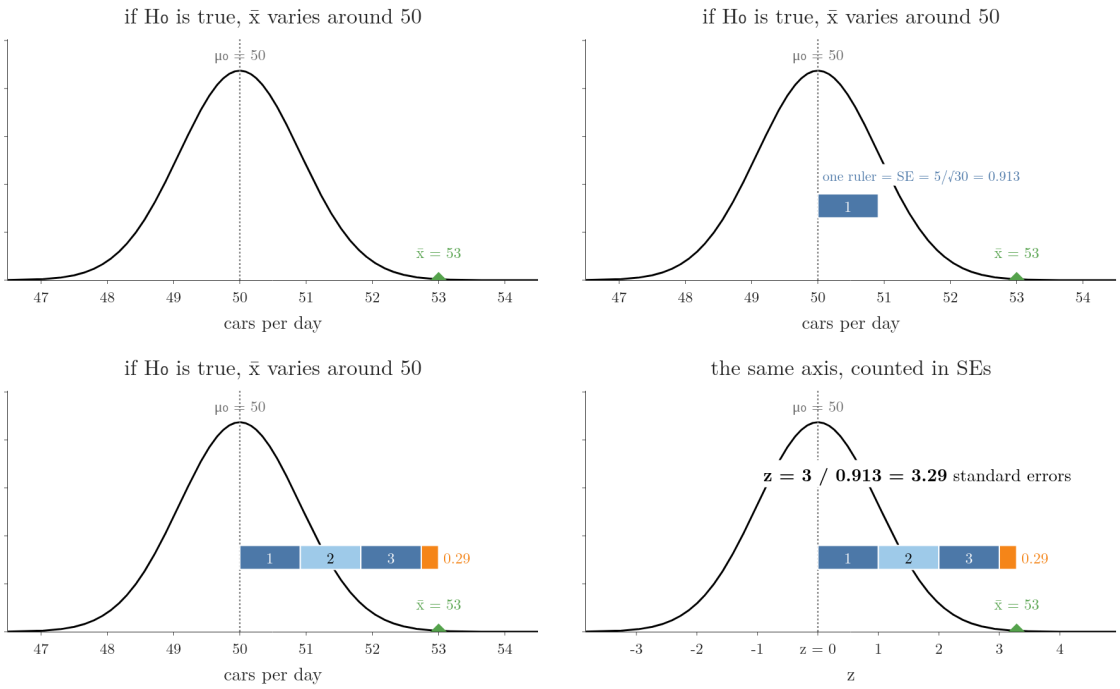


Figure 3: Counting standard errors: rulers of 0.913 laid from 50 towards 53 fit 3.29 times, so $z = 3.29$

Figure 3 does the same count for the factory: standard-error rulers laid end to end from $\mu_0 = 50$ to $\bar{x} = 53$. Watch the last frame: relabelling the same axis in rulers turns 53 into $z = 3.29$.

4.3 Assumptions

The one-sample **z-test** (G-2143) can be used when:

1. **Normality:** the population is normal, or the sample has at least 30 values, so that by the central limit theorem $\bar{X}$ is approximately normal;
2. **Known σ:** the population standard deviation is known;
3. **Random sample:** the sample is drawn at random.

If σ is unknown, the exact test is the t-test (see the one-sample t-test). The rats break assumption 2: we used s for σ. With $n = 100$ the difference is small. The area beyond ± 3 is 0.0027 on the normal curve and 0.0034 on the t curve with 99 degrees of freedom (notebook, section 5), and both lead to the same decision below.

5. How rare is too rare: the significance level

> **Key point:** α is the probability of rejecting H_0 when it is actually true; we fix it before the test, usually at 0.05.

How rare is our rat sample? On a normal curve, 99.7% of the area lies within 3 standard deviations of the centre (the **68-95-99.7 rule**, G-53). So only about 0.3% of samples land 3 or more standard errors away, on either side. The exact area is 0.27%, or 0.135% in each tail (Figure 2, second step). If H_0 were true, a sample as extreme as ours would turn up less than 1 time in 300.

Where to draw the line. "Less than 1 in 300" sounds rare. But we need a fixed rule that says how rare is rare enough to reject H_0. That threshold is the **significance level** (G-1801), written α:

> α is the probability of rejecting the **null hypothesis** (G-1361) when it is actually true.

With $\alpha = 0.05$, if we ran 100 tests in which H_0 was in fact true, about 5 would still reject it. This mistake is called a **Type I error** (G-2032) (see **Type I error** (G-2032)).

Think of α as the sensitivity setting of a smoke alarm. A very sensitive alarm (large α) goes off for burnt toast as well as for real fires; a dull one (small α) rarely gives false alarms but needs more smoke before it sounds.

The usual choices are 0.05 (5%) and 0.01 (1%). For most problems 0.05 works well and is treated as the standard. In a specialised field, a domain expert may choose another value from knowledge of the domain.

The significance level and the confidence level are related but not the same. A 95% confidence level corresponds to $\alpha = 0.05$:

$$\text{confidence level} = 1 - \alpha$$

(see the z-procedure formula).

Fix α first. Without α there is no boundary between "reject" and "fail to reject". Choosing it after seeing the result would let us reach either answer: for any observed z other than 0, a large enough α puts it in the rejection region. For the chips example of section 9 ($z = -1.58$, two-tailed), any α above this value would reject H_0:

$$2 \times P(Z > 1.58) = 0.114$$

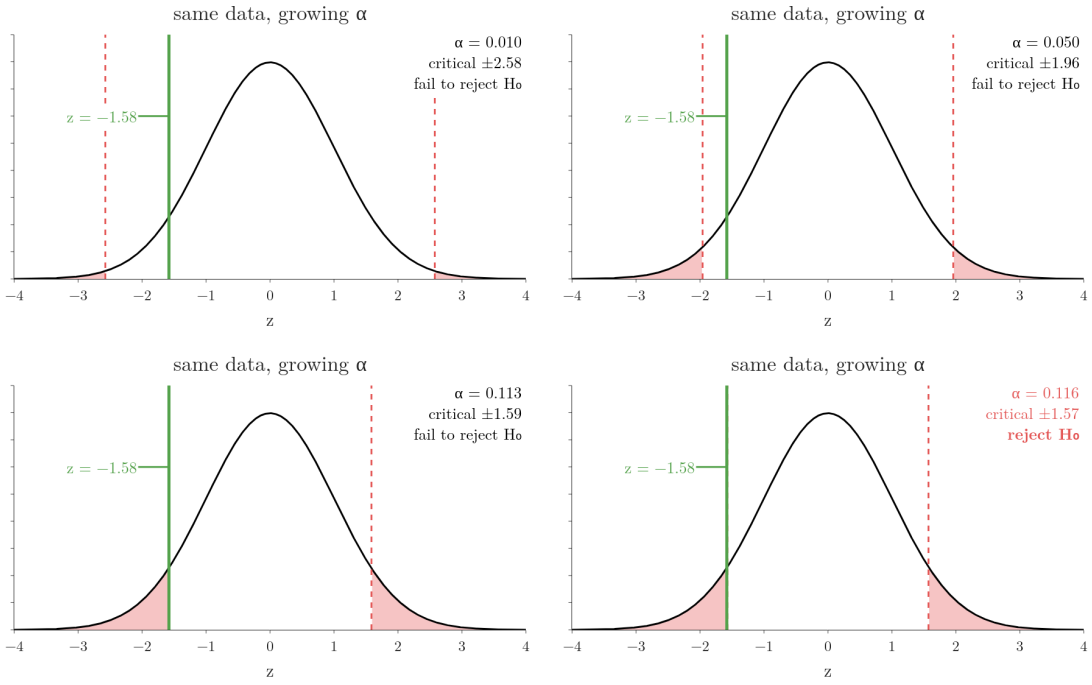


Figure 4: The chips $z = -1.58$ stays put while α grows; once α passes 0.114 the red tails reach it and the same data "rejects" H_0

In Figure 4, watch the red tails creep inward as α grows: nothing about the data changes, yet the verdict flips from "fail to reject" to "reject" just past $\alpha = 0.114$.

6. The rejection region and the critical value

> **Key point:** The rejection region is the tail of the standard normal curve with area α, on the side that H_1 points to; its boundary is the critical value.

The idea. Instead of asking "how rare is our sample?" each time, we mark in advance the part of the curve where samples are rarer than α allows. If our z lands there, we reject H_0.

The terms. The **rejection region** (also called the **critical region**) (G-1662) is the set of values of the **test statistic** (G-1963) for which we reject H_0. The rejection region lies in the tail or tails that H_1 points to, and its area is α. The rest of the curve is the region where we fail to reject H_0. The **critical value** (G-504) is the boundary of the rejection region:

- H_1 **with** $>$ **(right-tailed):** all of α in the right tail. For $\alpha = 0.05$ we need the z with area 0.95 to its left. The z-table has 0.9495 at 1.64 and 0.9505 at 1.65, so the critical value lies between them: 1.645.
- H_1 **with** $<$ **(left-tailed):** all of α in the left tail: -1.645.
- H_1 **with** $\neq$ **(two-tailed):** $\alpha/2 = 0.025$ in each tail: ± 1.96.

The decision rule is simple: if the test statistic falls in the rejection region, reject H_0; otherwise, fail to reject it. Carrying out a test this way is the **rejection region approach** (G-1663).

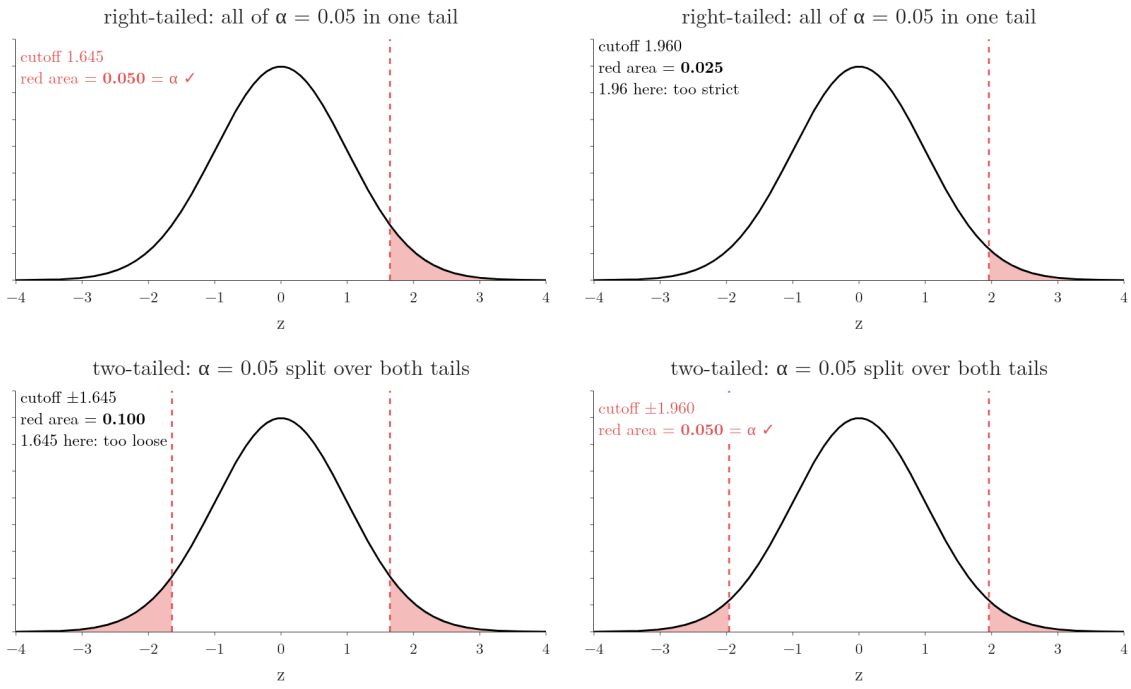


Figure 5: Finding the critical value: a cutoff slides until the red tail area equals $\alpha = 0.05$, at 1.645 for one tail and ± 1.96 for two

Figure 5 finds both critical values by sliding a cutoff until the red area is 0.05. Watch the middle frames: 1.96 leaves only 0.025 in one tail, and ± 1.645 leaves 0.10 across two.

Back to the rats. H_1 is $\mu \neq 1.2$, so the test is two-tailed with critical values ± 1.96. Our $z = -3$ is below -1.96: it falls in the rejection region, and we **reject** H_0. In seconds, the cutoffs are $1.2 \pm 1.96 \times 0.05$, that is 1.102 s and 1.298 s, and 1.05 s lies beyond the lower one (Figure 2, third step). The data gives strong evidence that the drug changes the response time.

Extra: One-sided and two-sided critical values (G-1383) are easy to mix up. At $\alpha = 0.05$, a **one-tailed test** (G-1385) uses 1.645 and a two-tailed test uses 1.96 (see reading a t-table for the same point with t). Using 1.96 for a one-tailed test makes it stricter than intended (a real α of 0.025); using 1.645 for a **two-tailed test** (G-2028) makes it looser (a real α of 0.10).

Python: Critical values come from `ppf`, the inverse of the CDF.

```
from scipy import stats

stats.norm.ppf(0.95)      # 1.645: right-tailed, alpha = 0.05
stats.norm.ppf(0.05)      # -1.645: left-tailed
stats.norm.ppf(0.975)     # 1.960: two-tailed, alpha = 0.05
```

7. One tail or two: the same data, two questions

> **Key point:** H_1 with $\neq$ puts the rejection region in both tails; H_1 with $<$ or $>$ puts all of it in one tail.

The idea. In the rat test, a very slow mean would also have counted against H_0: any effect, up or down, was of interest. So we looked at both tails. That is a **two-tailed test** (G-2028).

Suppose instead that, before collecting data, we had reason to expect the drug to *lower* response time, and only a lowering mattered. Then

$$H_0 : \mu = 1.2$$

$$H_1 : \mu < 1.2$$

and only very low sample means count against H_0. All of α goes into the left tail. That is a **one-tailed test** (G-1385), here a left-tailed one (G-1693).

The same rats, one tail. The **critical value** (G-504) is now -1.645, and in seconds the cutoff is:

$$1.2 - 1.645 \times 0.05 = 1.118 \text{ s}$$

Our $z = -3$ is below -1.645: reject H_0 (Figure 2, last step). The tail area beyond our sample is now one tail only: half of 0.27%, which is 0.135%.

	Two-tailed	One-tailed (left)
H_1	$\mu \neq 1.2$	$\mu < 1.2$
Critical value at $\alpha = 0.05$	± 1.96	-1.645
Cutoff in seconds	1.102 and 1.298	1.118
Area beyond our sample	0.27%	0.135%
Decision for $z = -3$	reject	reject

The direction of H_1 must be chosen before looking at the data; one-tailed and two-tailed tests show what goes wrong when it is not.

8. Example: did a training program raise productivity?

> **Key point:** $z = 3.29$ lies beyond the **critical value** (G-504) 1.645, so we reject H_0: the training program raised mean productivity.

In a car factory, an employee produces **50 units per day** on average, with a known population standard deviation of **5 units**: some employees make 45, some 55. To raise output, the company runs a training program for its employees. Afterwards it measures the productivity of a random sample of **30** employees and finds a sample mean of **53 units per day**. Has the training significantly increased productivity?

1. **Hypotheses.** No benefit, against an increase:

$$H_0 : \mu = 50$$

$$H_1 : \mu > 50$$

2. **Significance level.** $\alpha = 0.05$.
3. **Assumptions.** $n = 30$, so by the central limit theorem $\bar{X}$ is approximately normal. $\sigma = 5$ is known.
4. **Test.** Normal and σ known: the one-sample **z-test** (G-2143).
5. **Test statistic.** z.
6. **Conduct the test.** As in section 4.2:

$$z = \frac{53 - 50}{5/\sqrt{30}} = 3.29$$

7. **Decide.** H_1 uses $>$, so the rejection region is the right tail beyond 1.645. Since $3.29 > 1.645$, z falls in the rejection region (Figure 1, left): we **reject** H_0. The data gives strong evidence against H_0, in favour of H_1.
8. **Interpret.** The training program increased the mean daily productivity per employee above 50 units.

Python: The whole test in a few lines.

```
import numpy as np
from scipy import stats

z = (53 - 50) / (5 / np.sqrt(30))    # 3.29
z_crit = stats.norm.ppf(1 - 0.05)     # 1.645
z > z_crit                            # True: reject H0
```

9. Example: do chips packets weigh 50 g?

Key point: $z = -1.58$ lies between -1.96 and 1.96, so we fail to reject H_0: the sample does not show that the mean weight differs from 50 g.

A snack company claims that its packets of chips weigh **50 g** on average. A consumer watchdog wants to know whether the true mean weight differs from 50 g, more or less. It buys a random sample of **40** packets and finds a mean weight of **49 g**. The population standard deviation is known: **4 g**.

1. **Hypotheses.**

$$H_0 : \mu = 50$$

$$H_1 : \mu \neq 50$$

2. **Significance level.** $\alpha = 0.05$.
3. **Assumptions.** $n = 40 \geq 30$: $\bar{X}$ is approximately normal. $\sigma = 4$ is known.
4. **Test.** One-sample **z-test** (G-2143).
5. **Test statistic.** z.

6. **Conduct the test.**
$$z = \frac{49 - 50}{4/\sqrt{40}} = \frac{-1}{0.632} = -1.58$$
7. **Decide.** H_1 uses $\neq$: we do not know the direction, so the test is two-tailed. $\alpha = 0.05$ is split into 0.025 in each tail, giving **critical values** (G-504) ± 1.96. Since $-1.96 < -1.58 < 1.96$, z falls between them (Figure 1, right): we **fail to reject** H_0.
8. **Interpret.** The 40 packets do not give enough evidence that the mean weight differs from 50 g. The watchdog has no case against the company.

The same decision can be read in grams, as we did for the rats in seconds. Under H_0 the mean of 40 packets has this standard error:

$$4/\sqrt{40} = 0.632 \text{ g}$$

We fail to reject for any sample mean between a lower limit and an upper limit:

$$50 - 1.96 \times 0.632 = 48.76 \text{ g}$$
$$50 + 1.96 \times 0.632 = 51.24 \text{ g}$$

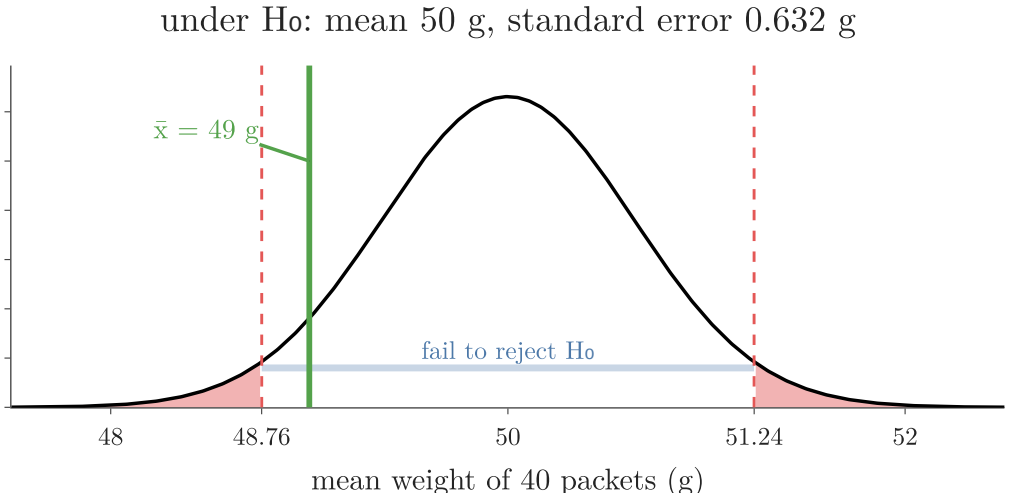


Figure 6: The chips test in grams: sample means between 48.76 g and 51.24 g fail to reject H_0; the watchdog's 49 g lies inside

In Figure 6, watch where 49 g sits: inside the blue band, one gram below the centre 50 g and only 0.24 g above the left cutoff 48.76 g.

Failing to reject does not prove that the packets weigh exactly 50 g on average. As failing to reject showed, failing to reject H_0 only means the evidence was not strong enough.

10. How the significance level moves the rejection region

> **Key point:** A smaller α pushes the **critical values** (G-504) outward and shrinks the rejection region, so rejecting a true H_0 becomes rarer.

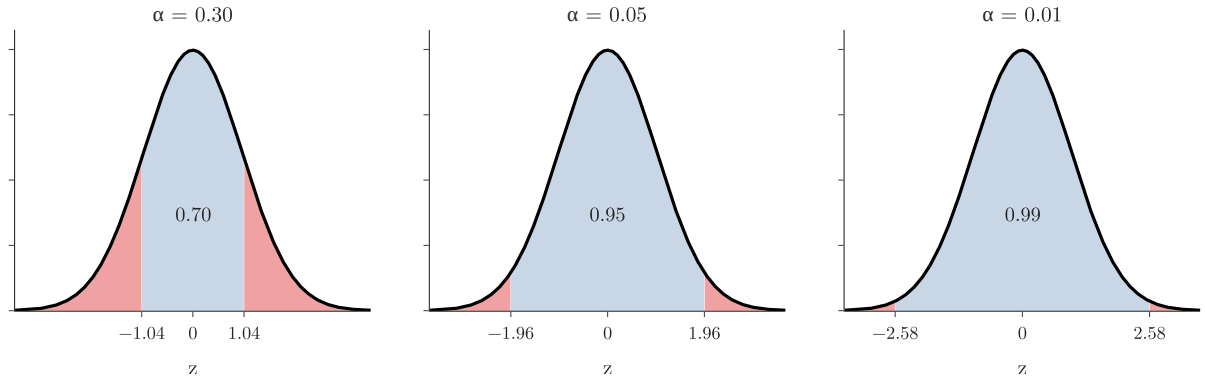


Figure 7: Two-tailed rejection regions (red) for three significance levels; the middle area is $1 - \alpha$

Figure 7 shows a two-tailed test at three **significance levels** (G-1801):

α	Area in each tail	Critical values
0.30	0.15	± 1.04
0.05	0.025	± 1.96
0.01	0.005	± 2.58

- **Lower** α (5% to 1%): the critical values move out and the region where we fail to reject grows. A true H_0 is rejected less often.
- **Higher** α (say 30%): the rejection region grows. Even when H_0 is true, the statistic now often lands in the rejection region by chance, and we wrongly reject H_0.

Figure 8 checks that meaning by simulation. We repeat the chips test of section 9 400 times with packets whose true mean really is 50 g, so H_0 is true every time. Watch each sample's z land under the curve: about 5% land in the red tails and wrongly reject H_0 (here 24 of 400, 6.0%). Then α sweeps over the same 400 values, and the rejected share follows it (124 of 400 at $\alpha = 0.30$). Last, the watchdog's own $z = -1.58$ lands between the critical values.

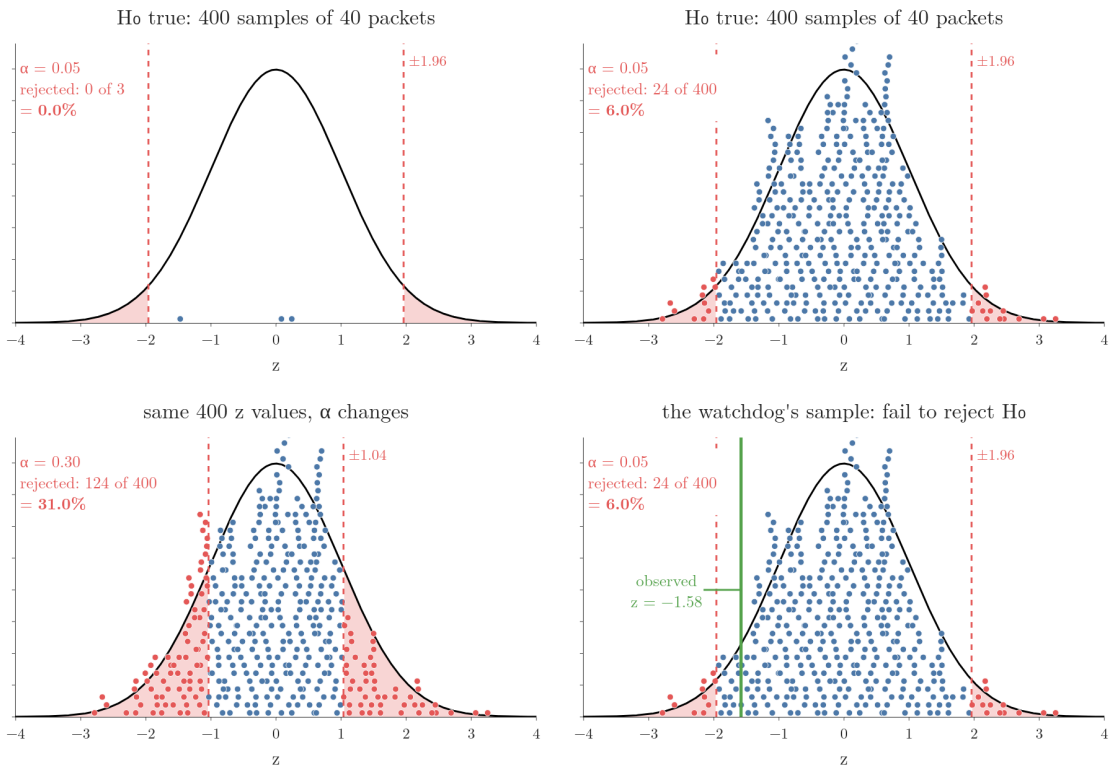


Figure 8: 400 z values from samples where H_0 is true drop onto the standard normal curve; red dots fall in the rejection region, and their share tracks α as α changes

The red area is exactly what "the probability of rejecting H_0 when it is actually true" means. A lower α has a cost too, which the trade-off between Type I and Type II errors explains.

11. The weakness of the rejection region approach

> **Key point:** The **rejection region approach** (G-1663) only says on which side of a boundary the statistic fell; it cannot say how strong the evidence is.

The rejection region approach gives a yes-or-no answer, and that causes two problems.

A knife-edge boundary. In a **two-tailed test** (G-2028) at $\alpha = 0.05$, $z = 1.97$ means "reject" and $z = 1.95$ means "fail to reject". A difference of 0.02 flips the decision, although the two samples are almost equally convincing.

No strength of evidence (G-1898). $z = 2$ and $z = 15$ both fall in the rejection region and both lead to "reject H_0". But they are very different:

z	Probability of a value at least this large if H_0 is true	Evidence against H_0
2	0.023	moderate
15	about 4×10^{-51}	overwhelming

A z of 15 is so unlikely under H_0 that it leaves almost no doubt, while a z of 2 is the kind of value that turns up 2 or 3 times in 100 by chance. The rejection region approach treats them the same.

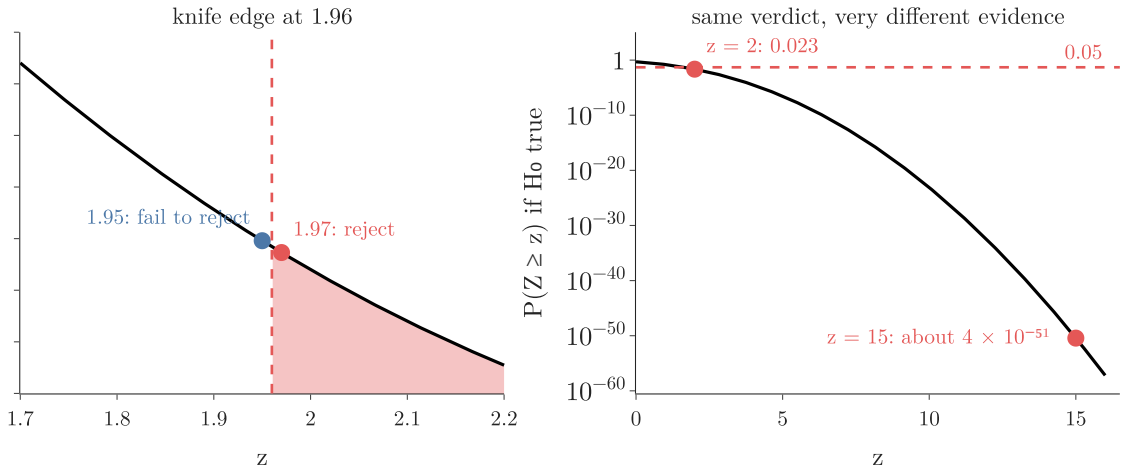


Figure 9: Left: $z = 1.95$ and $z = 1.97$ straddle 1.96 and get opposite verdicts. Right: $P(Z \geq z)$ on a log scale; $z = 2$ and $z = 15$ are both below 0.05, about 49 powers of ten apart

Figure 9 shows both problems. The left panel is the right tail of the standard normal curve, with the rejection region shaded. The right panel's vertical axis is a log scale: each gridline is 10^{10} times smaller than the one above it, so the tiny value at $z = 15$ still fits on the chart. Watch the right panel: the yes/no rule sees only that both points sit below the dashed 0.05 line.

The fix is the number we already met for the rats in section 5: the tail area beyond our own sample, 0.27% two-tailed. That area is the **p-value** (G-1433). It measures the strength of the evidence against H_0. The p-value approach is the one used in practice, and the subject of the definition of a p-value.

12. Summary

	Rat drug	Training program	Chips packets
H_0 / H_1	$\mu = 1.2$ / $\mu \neq 1.2$	$\mu = 50$ / $\mu > 50$	$\mu = 50$ / $\mu \neq 50$
Data	$n = 100$, $\bar{x} = 1.05$, $s = 0.5$	$n = 30$, $\bar{x} = 53$, $\sigma = 5$	$n = 40$, $\bar{x} = 49$, $\sigma = 4$
Tails	two	right	two
Critical value ($\alpha = 0.05$)	± 1.96	1.645	± 1.96
z	-3	3.29	-1.58
Decision	reject H_0	reject H_0	fail to reject H_0

- In each column the decision comes from comparing z with the **critical value** (G-504): -3 and 3.29 lie beyond theirs, so H_0 is rejected; -1.58 lies between -1.96 and 1.96, so it is not.
- The logic: assume H_0, find where sample means would land, and reject if ours lands somewhere very unlikely, because a sample that rare is hard to explain if H_0 were true.
- The statistic z counts standard errors:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

  We divide by $\sigma/\sqrt{n}$ because a sample mean spreads by the standard error, not by σ. The one-sample **z-test** (G-2143) needs normality (or $n \geq 30$), a known σ and a random sample; with σ unknown the exact test is the t-test.
- α, fixed in advance, is the probability of rejecting a true H_0; 0.05 is the standard choice. It is fixed first because choosing it after the result could reach either verdict (the chips data "rejects" once α passes 0.114).
- The rejection region has area α in the tail(s) H_1 points to; its boundary is the critical value, so the decision is one comparison of z with that boundary.
- Right-tailed 1.645, two-tailed ± 1.96 at $\alpha = 0.05$, because two tails split α into 0.025 each; mixing them up makes the test stricter or looser than intended.
- A smaller α shrinks the rejection region, so a true H_0 is rejected less often.
- The approach cannot tell weak from overwhelming evidence, because it only says on which side of the boundary z fell ($z = 1.97$ rejects, 1.95 does not); **p-values** (G-1433) can.
- This answers the opening question: assume H_0, measure in standard errors how far the sample lands from it, and reject when it lands in a tail region of area α.

13. Sources

Built from

- CampusX, "Session 45 - Hypothesis Testing Part 1 | DSMP 2023", YouTube, https://www.youtube.com/watch?v=S94mx6OL7kM. The eight steps, the significance level, the rejection region and the two worked examples (sections 5, 6, 8 and 9).

- Khan Academy, "Hypothesis testing and p-values", YouTube, https://www.youtube.com/watch?v=-FtlH4svqx4. The rat drug test and its logic (sections 3 to 6).
- Khan Academy, "One-tailed and two-tailed tests", YouTube, https://www.youtube.com/watch?v=mvy The same rat data as a one-tailed test (section 7).

14. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Rejection region approach (G-1663)	Carrying out a test by checking whether the test statistic falls beyond a fixed boundary.
Significance level (α) (G-1801)	The chance, fixed before the test, of wrongly declaring an effect when there is none (rejecting H_0 when it is actually true); usually 0.05.
Z-test (one-sample) (G-2143)	A test of whether a population's mean equals a claimed value μ_0, used when the population spread σ is known and $\bar{X}$ is normal: $z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$.
Z statistic (G-2137)	The number a z-test computes from the sample: how many standard errors the sample mean lies from the mean claimed by H_0.
Rejection region (critical region) (G-1662)	The values of the test statistic for which we reject H_0; its area under H_0 is α.
One-sided and two-sided critical value (G-1383)	The cut-off beyond which the test statistic leads to rejecting H_0 (the critical value): one-sided puts the whole α in one tail, two-sided puts $\alpha/2$ in each of the two tails.
Strength of evidence (G-1898)	How strongly the data speaks against the starting claim of no effect (H_0); the rejection region approach does not measure it.
Null hypothesis (H_0) (G-1361)	The statement of no effect, no difference or no relationship; assumed true until the data gives strong evidence against it.
Alternative hypothesis (H_1, H_a) (G-193)	The claim of an effect, difference or relationship: the statement that contradicts the null hypothesis H_0.

Term	Meaning
Critical value (G-504)	The cut-off on a z (or t) curve that leaves $\alpha/2$ in each tail; a confidence interval reaches this many standard errors either side of the estimate. It is 1.96 for 95% on the standard normal curve.
P-value (G-1433)	The probability, assuming H_0 is true, of getting a sample as or more extreme than ours.
Standard error (G-1872)	How much a statistic, such as the sample mean, changes from one sample to the next, so how precise it is as an estimate. It is the standard deviation of the sampling distribution; for the mean, $\sigma/\sqrt{n}$.
Sampling distribution of the sample mean (G-1735)	The distribution of the means of many samples of size n; by the central limit theorem it is close to normal, centred on the population mean, with spread $\sigma/\sqrt{n}$.
Test statistic (G-1963)	The number a test computes from the sample to make its decision, such as z or t.
68-95-99.7 rule (empirical rule) (G-53)	In a column shaped like a bell curve (a normal distribution), about 68.3%, 95.4% and 99.7% of values lie within 1, 2 and 3 standard deviations of the mean.
Type I error (false positive) (G-2032)	A false alarm: the test declares an effect when there is none (rejecting H_0 when it is actually true); its probability is α.
One-tailed test (one-sided test) (G-1385)	A test whose H_1 has a direction ($>$ or $<$), with the whole rejection region in one tail.
Two-tailed test (two-sided test) (G-2028)	A test that looks for an effect in either direction: H_1 uses $\neq$, and the significance level is split into $\alpha/2$ in each tail.
Right-tailed and left-tailed test (G-1693)	One-tailed tests that count only one tail as evidence against H_0: a right-tailed test rejects for large values of the test statistic (as for $H_1 : \mu > \mu_0$), a left-tailed test for small ones (as for $H_1 : \mu < \mu_0$).